Habib University - City Campus
Course Title: CS/CE 224/272 Object Oriented Programming and Design Methodologies
Section : L4
Instructor's names: Nadia Nasir

Examination: Class Quiz 02
Total Marks: 37 marks

Exam Date: 6th Oct 2025
Time Limit: 40 mins

ID: ____________________

Name: __________________

Q1: [07 pts] Fill the table below:

| Assume the following declarations:<br>int m = 10; // Assumed to be at address 20300<br>int n = 20; // Assumed to be at address 20304<br>int* p = &m; | | |
|---|---|---|
| Expression | Value | Comment |
| 1. `p` | 20300 | |
| 2. `*p` | 10 | |
| 3. `&n` | 20304 | |
| 4. `p = &n;` | p = 20304 | |
| 5. `++*p` | Value of n = 21 | |
| 6. `m = *p;` | m = 21 | |
| 7. `m = &p;` | Error | m is an int ,cannot assigned address(int*) to int |

Q2:[07 pts] Suppose the array primes, defined as

double primes[] = { 2, 3, 5, 7, 11, 13 };

starts at memory location 20300. What are the values of?

| | |
|---|---|
| a. primes | 20300 |
| b. *primes | 2 |
| c. primes + 4 | 20300+ (4*8) = 20332 |
| d. *(primes + 4) | 11 |
| e. primes[4] | 11 |
| f. &primes[4] | 20332 |

| | |
|---|---|
| g. primes+=1 | Error // cannot change the address in primes<br>Arrays are dedicated pointers |

Q3:[02 pts] Consider this program segment:

```
char a[] = "Mary had a little lamb";

char* p = a;

int count = 0;

while (*p != '\0')

{

 count++;

 while (*p != ' ' && *p != '\0') { p++; }

 while (*p == ' ') { p++; }

}
```

What is the value of count at the end of the outer while loop?

```
Ans:
5
Since there are 5 words in the array a.
```

Q4:[02 pts] Consider this function

```
int* grow(int a[], int size)

{

 int* result = new int[2 * size];

 for (int i = 0; i < size; i++) { result[i] = a[i]; }

 for (int i = size; i < 2 * size; i++) { result[i] = 0; }

 return result;

}
```

What are the contents of the array to which ***p*** points after the following statements?

```
int primes[] = { 2, 3, 5, 7, 11 };

int* p = grow(primes, 5);
```

Ans:
2 3 5 7 11 0 0 0 0 0

Q5:[12 pts] Explain the errors in the following statements. Not all lines contain errors. Each depends on the lines preceding it. Watch out for uninitialized pointers, NULL pointers, pointers to deleted objects, and confusing pointers with objects.

| | | |
|---|---|---|
| 1 | int* p = new int; | no error |
| 2 | p = 5; | Error.value of type int cannot assigned to value of type int* |
| 3 | *p = *p + 5; | after commenting line 2.<br>garbage value at *p will be stored |
| 4 | int* a = new int[10]; | array of 10 int will be allocated at heap |
| 5 | *a = 5; | value at index 0 = 5 // a[0]=5 |
| 6 | a[10] = 5; | value at index 10 = 5 // no array bound checking |
| 7 | delete a; | only release/delete first int in array a. |
| 8 | int* q; | no error |
| 9 | *q = 5; | Segmentation fault :no address for dereferencing |
| 10 | q = p; | q is pointing the same int in heap where p is pointing |
| 11 | delete q; | memory released |
| 12 | delete p; | The same memory block was deleted twice. |

Q6: [01 pts] Which of the following function declarations represents valid function overloading in C++?

```
void display(int x);

void display(double x);

void display(int x, int y);

void display(int t);
```

A) All functions are valid overloads
B) The last function is not a valid overload

D) Only the first two functions are valid overloads

Q7:[01 pts ]When functions are overloaded in C++, the version of the function that will be executed is determined at _____.

1. Compile Time
2. Run Time

Q8: [05 pts] Are both these functions valid if they are designed to accept an array as a parameter? Briefly explain:

```
#include <iostream>
using namespace std;
void foo(int arr[], int len){
    cout <<" Function 1 \n";
}
void foo(int* arr, int len){
    cout <<"Function 2 \n";
}
int main()
{
    int num[]= {45,66,4,2};
    foo(num , 4);

    return 0;
}
```

Ans:

```
horse.cpp:6:6: error: redefinition of 'void foo(int*, int)'
 void foo(int* arr, int len){
      ^~~
horse.cpp:3:6: note: 'void foo(int*, int)' previously defined here
 void foo(int arr[], int len){
      ^
```